VIVEKANANDHA SAINIK COACHING CENTER

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ALL INDIA SAINIK SCHOOLS ENTRANCE EXAMINATION (AISSEE) — CLASS VI

MOCK QUESTION PAPER — 02

DURATION: 150 MINUTES (2½ HOURS) | MAXIMUM MARKS: 300

Candidate Name: _____ **Roll Number:** _____ **OMR Sheet No.:** _____

PART A — EXAMINATION SCHEME & CANDIDATE INSTRUCTIONS

1. This booklet contains 125 Multiple Choice Questions divided into four sections: **Section A: Language (English)** (Q.1–25, 2 Marks each = 50 Marks); **Section B: Mathematics** (Q.26–75, 3 Marks each = 150 Marks); **Section C: Intelligence** (Q.76–100, 2 Marks each = 50 Marks); **Section D: General Knowledge** (Q.101–125, 2 Marks each = 50 Marks).
2. All questions are compulsory. There is **no negative marking** for wrong answers.
3. Darken only one circle on your OMR Answer Sheet using a Black or Blue ballpoint pen.

SECTION A : ENGLISH LANGUAGE (Q.1 TO 25)

Q.1. Choose the correct collective noun: A _____ of wolves was heard howling in the dark mountains.

(A) herd (B) pack
(C) flock (D) pride

Q.2. Fill in the blank with correct preposition: The old man divided his property _____ his three sons.

(A) between (B) among
(C) with (D) into

Q.3. Choose the correct article: Honest citizens always speak _____ truth.

(A) a (B) an
(C) the (D) no article

Q.4. Choose the correct reflexive pronoun: The young girl hurt _____ while skating on the ice.

(A) himself (B) herself
(C) itself (D) yourself

Q.5. Fill in the blank with superlative adjective: Iron is the _____ of all useful metals.

(A) useful (B) more useful
(C) most useful (D) as useful

Q.6. Choose the correct tense form: By this time tomorrow, we _____ reached New Delhi.

(A) will have (B) had
(C) have (D) were

Q.7. Identify the correctly spelt word from the given options:

(A) Necessary (B) Necesary
(C) Neccessary (D) Necessery

Q.8. Choose the word nearest in meaning (Synonym) to ABUNDANT:

(A) Plentiful (B) Scarce
(C) Small (D) Rare

Q.9. Choose the word opposite in meaning (Antonym) to DISCORD:

(A) Conflict (B) Harmony
(C) Noise (D) Harsh

Q.10. What is the meaning of the idiom 'A blessing in disguise'?

(A) A bad event that turns out to have good results (B) A terrible curse
(C) A fake reward (D) A secret plan

Q.11. Select the correct question tag: She dances gracefully on stage, _____?

(A) isn't she (B) doesn't she
(C) don't she (D) won't she

Q.12. What is the masculine gender form of 'Vixen'?

(A) Stallion (B) Fox

(C) Drake (D) Gander

Q.13. Select the correct plural form of 'Criterion':

(A) Criterions (B) Criteria

(C) Criterias (D) Criteri

Q.14. In the sentence 'The tired baby slept peacefully', the verb 'slept' is:

(A) Transitive (B) Intransitive

(C) Auxiliary (D) Modal

Q.15. In 'He rarely comes late to school', identify the type of adverb ('rarely'):

(A) Adverb of Frequency (B) Adverb of Manner

(C) Adverb of Place (D) Adverb of Time

Q.16. Choose the suitable conjunction: He ran fast _____ he missed the morning train.

(A) and (B) but

(C) so (D) because

Q.17. Rearrange the jumbled words: P: Honesty / Q: policy / R: is / S: the best

(A) PRSQ (B) PQSR

(C) PRQS (D) QSRP

Q.18. Choose the one word substitute for: 'A place where birds are housed and kept':

(A) Apiary (B) Aviary

(C) Aquarium (D) Zoo

Q.19. Fill in the blank with the appropriate interjection: '_____! We won the inter-school trophy!'

(A) Alas (B) Hurrah

(C) Hush (D) Fie

Q.20. Choose the correct word: The School Management gave its _____ to build a new library.

(A) assent (B) ascent

(C) accent (D) asset

READING COMPREHENSION PASSAGE

Read the following passage carefully and answer questions 21 to 25: The Gir National Park and Wildlife Sanctuary in Gujarat is the sole refuge and remaining natural habitat of the Asiatic Lion in the world. Established in 1965, the protected sanctuary covers an area of nearly 1,412 square kilometers. The ecosystem of Gir consists of dry deciduous teak forests, open grasslands, and thorn scrub. Apart from Asiatic lions, the park is inhabited by leopards, jungle cats, hyenas, sambar deer, chital, and over 300 species of birds. The Maldhari community has coexisted with the wildlife in Gir for generations, grazing their livestock within the forest margins while respecting nature.

Q.21. According to the passage, in which state is Gir National Park situated?

(A) Rajasthan (B) Gujarat

(C) Madhya Pradesh (D) Maharashtra

Q.22. Which iconic wild animal finds its sole natural habitat in Gir forest?

(A) Bengal Tiger (B) Asiatic Lion

(C) Snow Leopard (D) One-horned Rhinoceros

Q.23. In which year was the Gir wildlife sanctuary established?

(A) 1955 (B) 1965

(C) 1975 (D) 1985

Q.24. Which traditional community coexists with the wildlife inside Gir forest?

(A) Toda community (B) Maldhari community

(C) Santhal community (D) Bhil community

Q.25. Which word in the passage means 'a place providing shelter or safety'?

(A) Ecosystem (B) Refuge

(C) Scrub (D) Species

SECTION B : MATHEMATICS (Q.26 TO 50)

Q.26. Find the sum of all prime factors of 210:

(A) 15 (B) 17

(C) 19 (D) 21

Q.27. Find the product of the place value of 5 in 45,382 and place value of 3 in 2,341:

(A) 1,50,000 (B) 15,00,000

(C) 15,000 (D) 1,50,00,000

Q.28. Solve the Roman Numeral calculation: CXCV – LXIV

(A) CXXI (B) CXXXI

(C) CXLI (D) CXXIX

Q.29. Find the smallest 4-digit number which is exactly divisible by 12, 15, and 18:

(A) 1020 (B) 1080

(C) 1140 (D) 1000

Q.30. The HCF and LCM of two numbers are 16 and 240 respectively. If one number is 48, find the other number:

(A) 60 (B) 80

(C) 90 (D) 72

Q.31. Simplify using BODMAS: $90 - [35 - \{42 \div (10 - 3) \times 2\}]$

(A) 67 (B) 73

(C) 55 (D) 61

Q.32. Calculate: $3\ 1/3 + 2\ 1/4 - 1\ 1/2$

(A) 4 1/12 (B) 3 11/12

(C) 4 5/12 (D) 3 7/12

Q.33. Find the difference between 5/8 of 64 and 3/7 of 49:

(A) 17 (B) 19

(C) 21 (D) 23

Q.34. Convert the decimal 3.125 into a mixed fraction in its lowest terms:

(A) 3 1/8 (B) 3 1/4

(C) 3 3/8 (D) 3 1/5

Q.35. A car covers a distance of 240 km in 4 hours. How long will it take to cover 360 km at the same speed?

(A) 5 hours (B) 6 hours

(C) 7 hours (D) 8 hours

Q.36. If $2A = 3B = 4C$, find the ratio A : B : C:

(A) 2 : 3 : 4 (B) 6 : 4 : 3

(C) 4 : 3 : 2 (D) 3 : 4 : 6

Q.37. A dealer bought a ceiling fan for ■ 1,200 and spent ■ 100 on transport. He sold it for ■ 1,430. Find his gain percentage:

(A) 8% (B) 10%

(C) 12% (D) 15%

Q.38. By selling a table for ■ 1,350, a trader loses 10%. At what price should he sell it to gain 10%?

(A) ■ 1,550 (B) ■ 1,650

(C) ■ 1,600 (D) ■ 1,700

Q.39. The average marks obtained by 5 students in a test are 78. If four students scored 72, 80, 85, and 68, what did the 5th student score?

(A) 82 (B) 85

(C) 88 (D) 90

Q.40. In an examination, 450 students appeared. If 90% passed the test, how many students failed?

(A) 35 (B) 45

(C) 50 (D) 40

Q.41. Find the total amount received when ■ 12,000 is invested at 8% per annum simple interest for 2.5 years:

(A) ■ 14,200 (B) ■ 14,400

(C) ■ 14,600 (D) ■ 13,800

Q.42. At what rate of simple interest per annum will ■ 6,000 yield an interest of ■ 1,440 in 3 years?

(A) 6% (B) 8%

(C) 9% (D) 10%

Q.43. An angle is equal to 3 times its supplement. Find the measure of the angle:

(C) 145° (D) 150°

Q.44. Two parallel lines are intersected by a transversal. If one interior angle on the same side is 65°, find the other interior angle:

(A) 105° (B) 115°

(C) 125° (D) 65°

Q.45. In a right-angled triangle, one acute angle measures 38°. Find the measure of the other acute angle:

(A) 42° (B) 52°

(C) 62° (D) 48°

Q.46. Find the area of a square playground whose perimeter is 64 metres:

(A) 225 m² (B) 256 m²

(C) 289 m² (D) 196 m²

Q.47. The length of a rectangular field is 35 m and its breadth is 20 m. Find the cost of leveling it at ■ 8 per m²:

(A) ■ 5,200 (B) ■ 5,600

(C) ■ 6,000 (D) ■ 4,800

Q.48. The diameter of a circular wheel is 70 cm. How far (in metres) will it travel in 100 complete revolutions?

(A) 200 m (B) 220 m

(C) 240 m (D) 180 m

Q.49. A cube has a total surface area of 294 cm². Find the volume of the cube:

(A) 216 cm³ (B) 343 cm³

(C) 512 cm³ (D) 294 cm³

Q.50. Find the capacity in litres of a rectangular water tank of dimensions 3 m × 2 m × 1.5 m:

(A) 8,000 L (B) 9,000 L

(C) 10,000 L (D) 7,500 L

Q.51. An express bus covers 180 km at a constant speed of 45 km/hr. How much time does it take?

(A) 3.5 hours (B) 4 hours

(C) 4.5 hours (D) 5 hours

Q.52. A sprinter runs a 400 m race in 50 seconds. Find his running speed in km/hr:

(A) 25.2 km/hr (B) 28.8 km/hr

(C) 30.0 km/hr (D) 32.4 km/hr

Q.53. Solve for y in the linear equation: $4(y - 3) = 2(y + 5)$

(A) 9 (B) 11

(C) 13 (D) 15

Q.54. If a = 4, b = 2, and c = 3, evaluate the algebraic expression: $(a^2 + b^2) \div (c + 1)$

(A) 4 (B) 5

(C) 6 (D) 8

Q.55. Two numbers differ by 18. If their sum is 84, find the smaller number:

(A) 31 (B) 33

(C) 51 (D) 35

Q.56. At what temperature numerical reading is exactly identical on both Celsius and Fahrenheit scales?

(A) –32° (B) –40°

(C) 0° (D) 100°

Q.57. Evaluate the integer expression: (+28) + (–42) – (–15) + (–11)

(A) –10 (B) +10

(C) –20 (D) 0

Q.58. Find the sum of all prime numbers less than 20:

(A) 75 (B) 77

(C) 80 (D) 72

Q.59. Which of the following numbers is divisible by both 4 and 8?

(A) 5,420 (B) 7,816

(C) 6,314 (D) 9,522

Q.60. Find the value of x if 3, 9, x, and 27 are in proportion:

(A) 6 (B) 9

(C) 12 (D) 15

Q.61. What is the perimeter of an equilateral triangle of side 14.2 cm?

(A) 40.6 cm (B) 42.6 cm

(C) 44.2 cm (D) 41.8 cm

Q.62. How many 500 mL water bottles can be filled from a drum containing 45 litres of water?

(A) 80 (B) 90

(C) 100 (D) 75

Q.63. Estimate the product 482 × 319 by rounding off each number to its nearest hundred:

(A) 1,20,000 (B) 1,50,000

(C) 1,60,000 (D) 1,40,000

Q.64. An angle measuring 89.5° is classified as a/an:

(A) Acute angle (B) Right angle

(C) Obtuse angle (D) Reflex angle

Q.65. How many lines of symmetry does a non-square rectangle have?

(A) 1 (B) 2

(C) 4 (D) Infinite

Q.66. The average height of 4 girls is 145 cm. If three girls are 140 cm, 148 cm, and 152 cm tall, find the height of the fourth girl:

(A) 138 cm (B) 140 cm

(C) 142 cm (D) 144 cm

Q.67. Express 45 paise as a percentage of ■ 3.00:

(A) 12% (B) 15%

(C) 18% (D) 20%

Q.68. The distance around the outer boundary of a circle is called its:

(A) Diameter (B) Circumference

(C) Radius (D) Chord

Q.69. Read the two mathematical statements: Statement I: The sum of any two odd numbers is always even. Statement II: The product of any two prime numbers is always prime.

(A) Both I and II are true (B) Both I and II are false

(C) Statement I is true, II is false (D) Statement I is false, II is true

Q.70. If HCF(A, B) = 12 and LCM(A, B) = 180, and A = 36, find the value of B:

(A) 48 (B) 60

(C) 72 (D) 54

Q.71. A furniture dealer buys 12 chairs at ■ 250 per chair and sells all of them for ■ 3,600. Find his profit percentage:

(A) 15% (B) 20%

(C) 25% (D) 18%

Q.72. Simplify the fractional expression: 7/10 + 2/5 − 3/20

(A) 17/20 (B) 19/20

(C) 21/20 (D) 4/5

Q.73. The ratio of ages of Father and Son is 7 : 2. If Father is 42 years old, how old is the Son?

(A) 10 years (B) 12 years

(C) 14 years (D) 16 years

Q.74. A cyclist travels 45 km in 3 hours. At the same speed, how much distance will he cover in 5 hours?

(A) 65 km (B) 75 km

(C) 80 km (D) 70 km

Q.75. In a bar graph, 1 cm on the vertical axis represents 20 students. A bar of height 4.5 cm represents how many students?

(A) 80 (B) 90

(C) 100 (D) 85

SECTION C : INTELLIGENCE (Q.76 TO 100)

Q.76. Find the missing term in the number series: 2, 6, 12, 20, 30, ?

(A) 40 (B) 42

(C) 44 (D) 46

Q.77. Complete the letter series in reverse order: Z, X, V, T, R, ?

(A) O (B) P

(C) Q (D) N

Q.78. Complete the analogy: Doctor : Hospital :: Teacher : ?

(A) Office (B) School

(C) College (D) Library

Q.79. Complete the number analogy: 8 : 64 :: 11 : ?

(A) 110 (B) 121

(C) 132 (D) 144

Q.80. Choose the odd one out from the given domestic and wild animals:

(A) Dog (B) Cat

(C) Lion (D) Cow

Q.81. Choose the odd number out from the given options:

(A) 49 (B) 81

(C) 100 (D) 125

Q.82. If 'LION' is coded as 'MJPO', how is 'TIGER' written in that same code language?

(A) UJHFS (B) VKIHT

(C) UJHES (D) TIGFR

Q.83. A man pointed to a woman and said, 'Her father is the only son of my father.' How is the woman related to the man?

(A) Sister (B) Daughter

(C) Mother (D) Aunt

Q.84. Rohan walks 6 km North, turns East and walks 8 km. How far is he from his starting point in a direct line?

(A) 12 km (B) 10 km

(C) 14 km (D) 8 km

Q.85. In a row of 30 cadets, Rahul is 15th from the right end. What is his position from the left end?

(A) 15th (B) 16th

(C) 17th (D) 14th

Q.86. Arrange the words in alphabetical dictionary order: 1. Paper 2. Package 3. Palace 4. Page 5. Pain

(A) 2, 4, 5, 3, 1 (B) 2, 5, 4, 3, 1

(C) 4, 2, 5, 3, 1 (D) 2, 4, 3, 5, 1

Q.87. Arrange the units in ascending logical sequence: 1. Hour 2. Second 3. Day 4. Minute 5. Week

(A) 2, 4, 1, 3, 5 (B) 2, 1, 4, 3, 5

(C) 4, 2, 1, 3, 5 (D) 2, 4, 3, 1, 5

Q.88. If '+' means '–', '–' means '×', '×' means '÷', and '÷' means '+', evaluate: 20 – 4 × 2 + 10 ÷ 5

(A) 30 (B) 35

(C) 40 (D) 25

Q.89. Which word CAN be formed using the letters of 'COMPUTATION'?

(A) ACTION (B) STATION

(C) COMPUTER (D) NATION

Q.90. Find the missing number in the grid pattern: Row 1: 5, 4, 20 | Row 2: 6, 7, 42 | Row 3: 8, 9, ?

(A) 64 (B) 72

(C) 81 (D) 56

Q.91. Choose the correct mirror image of the digits '2 0 2 6' when mirror is placed on the right:

(A) 6 2 0 2 (flipped) (B) 2 0 2 6

(C) 6 0 2 2 (D) 2 2 0 6

Q.92. Which Venn diagram best represents: Engineers, Doctors, Human Beings?

(A) Two separate circles inside a large circle (B) Three intersecting circles

(C) Three concentric circles (D) Three separate circles

Q.93. How many total squares are present in a 3 × 3 grid?

(A) 9 (B) 14

(C) 12 (D) 16

Q.94. If 15th August of a non-leap year was a Friday, what day of the week was 26th August?

(A) Monday (B) Tuesday

(C) Wednesday (D) Sunday

Q.95. Which pair of words displays a different relationship compared to others?

(A) Sun : Light (B) Fire : Heat

(C) Ice : Cold (D) Moon : Star

Q.96. Given Statements: All roses are flowers. Some flowers fade quickly. Which conclusion is valid?

(A) All roses fade quickly (B) Some roses may fade quickly

(C) No rose ever fades (D) All flowers are roses

Q.97. Complete the geometric analogy: Circle : Perimeter :: Polygon : ?

(A) Area (B) Perimeter

(C) Diagonal (D) Volume

Q.98. If 'RED' = 27 (18 + 5 + 4), what is the alphabetical sum code for 'BLUE'?

(A) 36 (B) 40

(C) 42 (D) 38

Q.99. 4 boys P, Q, R, S sit in a circle facing center. P is left of Q, R is opposite P. Who is to immediate right of P?

(A) Q (B) R

(C) S (D) Cannot be determined

Q.100. A standard die has 6 faces numbered 1 to 6. Opposite faces sum to 7. What face is opposite to 3?

(A) 2 (B) 4

(C) 5 (D) 6

SECTION D : GENERAL KNOWLEDGE (Q.101 TO 125)

Q.101. Which scientific device is used to measure atmospheric pressure?

(A) Thermometer (B) Barometer

(C) Ammeter (D) Speedometer

Q.102. What is the official National Aquatic Animal of India?

(A) Blue Whale (B) Ganges River Dolphin

(C) Saltwater Crocodile (D) Green Sea Turtle

Q.103. Which classical dance form originates from the state of Tamil Nadu?

(A) Kathak (B) Bharatanatyam

(C) Kathakali (D) Odissi

Q.104. What is the equivalent rank in the Indian Air Force for a 'Major General' in the Indian Army?

(A) Air Commodore
(B) Air Vice Marshal
(C) Air Marshal
(D) Group Captain

Q.105. Which vitamin is synthesized naturally in human skin upon exposure to sunlight?

(A) Vitamin A
(B) Vitamin D
(C) Vitamin C
(D) Vitamin K

Q.106. Iodine deficiency in the human diet leads to swelling of the thyroid gland, known as:

(A) Anemia
(B) Goitre
(C) Rickets
(D) Scurvy

Q.107. Which organ do snakes use primarily to sense vibrations from the ground?

(A) External ears
(B) Lower jaw / Skin
(C) Eyes
(D) Nostrils

Q.108. Pitcher plant and Sundew are specialized examples of:

(A) Parasitic plants
(B) Insectivorous plants
(C) Saprophytic plants
(D) Symbiotic plants

Q.109. What is a 'Johad', traditionally built in rural Rajasthan?

(A) Folk song
(B) Rainwater harvesting earthen check dam
(C) Tribal festival
(D) Harvest dance

Q.110. At what temperature does pure water freeze at standard atmospheric pressure?

(A) 0°C (32°F)
(B) 100°C
(C) −4°C
(D) 4°C

Q.111. Which is the largest planet in our Solar System?

(A) Saturn
(B) Jupiter
(C) Neptune
(D) Uranus

Q.112. Where is the famous Sun Temple located in Odisha?

(A) Puri
(B) Konark
(C) Bhubaneswar
(D) Cuttack

Q.113. Who composed the ancient epic 'Ramayana' in Sanskrit language?

(A) Maharshi Valmiki
(B) Ved Vyas
(C) Kalidasa
(D) Tulsidas

Q.114. Which branch of the Indian Armed Forces operates the aircraft carrier INS Vikramaditya?

(A) Indian Army
(B) Indian Navy
(C) Indian Air Force
(D) Indian Coast Guard

Q.115. Which of the following is a clean renewable source of energy?

(A) Petroleum
(B) Wind energy
(C) Natural gas
(D) Coal

Q.116. The official headquarters of UNESCO is located in which city?

(A) Geneva
(B) Paris
(C) New York
(D) Vienna

Q.117. What is the young one of a Goose called?

(A) Cygnet
(B) Gosling
(C) Duckling
(D) Fawn

Q.118. Which physical process converts liquid water into water vapour upon heating?

(A) Condensation
(B) Evaporation
(C) Precipitation
(D) Freezing

Q.119. The Santosh Trophy is a major historic tournament associated with which sport in India?

(A) Hockey
(B) Football
(C) Cricket
(D) Basketball

Q.120. Which component of blood carries oxygen from lungs to all body cells?

(A) White Blood Cells
(B) Hemoglobin in Red Blood Cells
(C) Platelets
(D) Plasma

Q.121. Who was the first Indian to win an individual Gold Medal at the Olympic Games?

(A) Milkha Singh
(B) Abhinav Bindra
(C) Neeraj Chopra
(D) Sushil Kumar

Q.122. Which invisible force pulls objects toward the center of the Earth?

(A) Magnetic force
(B) Gravitational force
(C) Frictional force
(D) Electrostatic force

Q.123. Which state in India is the largest producer of Tea?

(A) West Bengal
(B) Assam
(C) Kerala
(D) Tamil Nadu

Q.124. The famous Jim Corbett National Park is located in which Indian state?

(A) Himachal Pradesh
(B) Uttarakhand
(C) Madhya Pradesh
(D) Rajasthan

Q.125. Who is widely regarded as the 'Father of the Indian Space Program'?

(A) Dr. APJ Abdul Kalam
(B) Dr. Vikram Sarabhai
(C) Dr. Homi Bhabha
(D) Satish Dhawan

PART B — ANSWER KEY (MOCK PAPER 02 : Q.1 TO 125)

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Q. No.	Opt	Answer	Q. No.	Opt	Answer	Q. No.	Opt	Answer	Q. No.	Opt	Answer
1	(B)	pack	33	(B)	19	65	(B)	2	97	(B)	Perimeter
2	(B)	among	34	(A)	3 1/8	66	(B)	140 cm	98	(B)	40
3	(C)	the	35	(B)	6 hours	67	(B)	15%	99	(A)	Q
4	(B)	herself	36	(B)	6 : 4 : 3	68	(B)	Circumference	100	(B)	4
5	(C)	most useful	37	(B)	10%	69	(C)	Statement I is t..	101	(B)	Barometer
6	(A)	will have	38	(B)	■ 1,650	70	(B)	60	102	(B)	Ganges River Dol..
7	(A)	Necessary	39	(B)	85	71	(B)	20%	103	(B)	Bharatanatyam
8	(A)	Plentiful	40	(B)	45	72	(B)	19/20	104	(B)	Air Vice Marshal
9	(B)	Harmony	41	(B)	■ 14,400	73	(B)	12 years	105	(B)	Vitamin D
10	(A)	A bad event that..	42	(B)	8%	74	(B)	75 km	106	(B)	Goitre
11	(B)	doesn't she	43	(B)	135°	75	(B)	90	107	(B)	Lower jaw / Skin
12	(B)	Fox	44	(B)	115°	76	(B)	42	108	(B)	Insectivorous pl..
13	(B)	Criteria	45	(B)	52°	77	(B)	P	109	(B)	Rainwater harves..
14	(B)	Intransitive	46	(B)	256 m²	78	(B)	School	110	(A)	0°C (32°F)
15	(A)	Adverb of Freque..	47	(B)	■ 5,600	79	(B)	121	111	(B)	Jupiter
16	(B)	but	48	(B)	220 m	80	(C)	Lion	112	(B)	Konark
17	(A)	PRSQ	49	(B)	343 cm³	81	(D)	125	113	(A)	Maharshi Valmiki
18	(B)	Aviary	50	(B)	9,000 L	82	(A)	UJHFS	114	(B)	Indian Navy
19	(B)	Hurrah	51	(B)	4 hours	83	(B)	Daughter	115	(B)	Wind energy
20	(A)	assent	52	(B)	28.8 km/hr	84	(B)	10 km	116	(B)	Paris
21	(B)	Gujarat	53	(B)	11	85	(B)	16th	117	(B)	Gosling
22	(B)	Asiatic Lion	54	(B)	5	86	(A)	2, 4, 5, 3, 1	118	(B)	Evaporation
23	(B)	1965	55	(B)	33	87	(A)	2, 4, 1, 3, 5	119	(B)	Football
24	(B)	Maldhari community	56	(B)	–40°	88	(B)	35	120	(B)	Hemoglobin in Re..
25	(B)	Refuge	57	(A)	–10	89	(A)	ACTION	121	(B)	Abhinav Bindra
26	(B)	17	58	(B)	77	90	(B)	72	122	(B)	Gravitational fo..
27	(B)	15,00,000	59	(B)	7,816	91	(A)	6 2 0 2 (flipped)	123	(B)	Assam
28	(B)	CXXXI	60	(B)	9	92	(A)	Two separate cir..	124	(B)	Uttarakhand
29	(B)	1080	61	(B)	42.6 cm	93	(B)	14	125	(B)	Dr. Vikram Sarab..
30	(B)	80	62	(B)	90	94	(B)	Tuesday			
31	(A)	67	63	(B)	1,50,000	95	(D)	Moon : Star			
32	(A)	4 1/12	64	(A)	Acute angle	96	(B)	Some roses may f..			

PART C — DETAILED STEP-BY-STEP SOLUTIONS (MOCK PAPER 02)

VIVEKANANDHA SAINIK COACHING CENTER

SOLUTIONS : SECTION A : ENGLISH LANGUAGE

Q.1. Correct Option: (B)
Explanation: A group of wolves is called a 'pack'. Therefore, option (B) is correct.

Q.2. Correct Option: (B)
Explanation: 'Among' is used when distributing or referring to more than two persons/things. Option (B) is correct.

Q.3. Correct Option: (C)
Explanation: 'The' is used before specific abstract nouns like 'the truth'. Option (C) is correct.

Q.4. Correct Option: (B)
Explanation: Reflexive pronoun for a female singular subject ('girl') is 'herself'. Option (B) is correct.

Q.5. Correct Option: (C)
Explanation: Superlative degree 'most useful' is used when comparing one item with all others in a group. Option (C) is correct.

Q.6. Correct Option: (A)
Explanation: Future Perfect tense ('will have reached') expresses an action completed by a certain future time. Option (A) is correct.

Q.7. Correct Option: (A)
Explanation: The correct spelling is 'Necessary' (one 'c' and double 's'). Option (A) is correct.

Q.8. Correct Option: (A)
Explanation: 'Abundant' means existing or available in large quantities; synonym is 'Plentiful' (A).

Q.9. Correct Option: (B)
Explanation: 'Discord' means disagreement between people; its antonym is 'Harmony' (B).

Q.10. Correct Option: (A)
Explanation: 'A blessing in disguise' refers to an apparent misfortune that eventually produces a good result (A).

Q.11. Correct Option: (B)
Explanation: Simple present positive sentence takes a negative auxiliary tag ('doesn't she?'). Option (B) is correct.

Q.12. Correct Option: (B)
Explanation: 'Fox' is the masculine form and 'Vixen' is the feminine form. Option (B) is correct.

Q.13. Correct Option: (B)
Explanation: Nouns of Greek origin ending in '-on' change to '-a' in plural ('Criteria'). Option (B) is correct.

Q.14. Correct Option: (B)
Explanation: 'Slept' does not require a direct object to complete its meaning, so it is an Intransitive Verb (B).

Q.15. Correct Option: (A)
Explanation: 'Rarely' shows how often an action occurs, so it is an Adverb of Frequency (A).

Q.16. Correct Option: (B)
Explanation: 'But' joins two contrasting clauses (running fast vs missing train). Option (B) is correct.

Q.17. Correct Option: (A)
Explanation: Correct order: P (Honesty) + R (is) + S (the best) + Q (policy) = PRSQ (Option A).

Q.18. Correct Option: (B)
Explanation: An 'Aviary' is a large cage or enclosure for keeping birds. Option (B) is correct.

Q.19. Correct Option: (B)
Explanation: 'Hurrah!' expresses joy, triumph, or exultation. Option (B) is correct.

Q.20. Correct Option: (A)
Explanation: 'Assent' means official agreement or approval. Option (A) is correct.

Q.21. Correct Option: (B)
Explanation: The passage states that Gir National Park is situated in Gujarat. Option (B) is correct.

Q.22. Correct Option: (B)
Explanation: The passage highlights Gir as the 'sole refuge of the Asiatic Lion in the world'. Option (B) is correct.

Q.23. Correct Option: (B)
Explanation: The passage states that the sanctuary was established in 1965. Option (B) is correct.

Q.24. Correct Option: (B)
Explanation: The passage explicitly names the 'Maldhari community'. Option (B) is correct.

Q.25. Correct Option: (B)
Explanation: 'Refuge' means a condition of being safe or a shelter from pursuit or danger. Option (B) is correct.

SOLUTIONS : SECTION B : MATHEMATICS

Q.26. Correct Option: (B)
Explanation: Prime factorization of 210 = 2 × 3 × 5 × 7. Sum = 2 + 3 + 5 + 7 = 17 (B).

Q.27. Correct Option: (B)
Explanation: Place value of 5 in 45,382 = 5,000. Place value of 3 in 2,341 = 300. Product = 5000 × 300 = 15,00,000 (B).

Q.28. Correct Option: (B)
Explanation: CXCV = 195. LXIV = 64. 195 – 64 = 131. In Roman numerals, 131 = CXXXI (B).

Q.29. Correct Option: (B)
Explanation: LCM(12, 15, 18) = 180. Smallest 4-digit number = 1000. 1000 ÷ 180 = 5 remainder 100. Next multiple = 180 × 6 = 1080 (B).

Q.30. Correct Option: (B)
Explanation: Other number = (HCF × LCM) / First number = (16 × 240) / 48 = 80 (B).

Q.31. Correct Option: (A)
Explanation: 10 – 3 = 7; 42 ÷ 7 = 6; 6 × 2 = 12. Then 35 – 12 = 23. Finally 90 – 23 = 67 (A).

Q.32. Correct Option: (A)
Explanation: (10/3) + (9/4) – (3/2) = LCM(12) = (40 + 27 – 18) / 12 = 49 / 12 = 4 1/12 (A).

Q.33. Correct Option: (B)
Explanation: (5/8 × 64) = 40. (3/7 × 49) = 21. Difference = 40 – 21 = 19 (B).

Q.34. Correct Option: (A)
Explanation: 3.125 = 3 + 125/1000 = 3 + 1/8 = 3 1/8 (A).

Q.35. Correct Option: (B)
Explanation: Speed = 240 / 4 = 60 km/hr. Time required = 360 / 60 = 6 hours (B).

Q.36. Correct Option: (B)
Explanation: A : B : C = (1/2) : (1/3) : (1/4). Multiplying by LCM (12) gives 6 : 4 : 3 (B).

Q.37. Correct Option: (B)
Explanation: Total Cost Price = 1200 + 100 = ■ 1,300. Selling Price = ■ 1,430. Gain = 130. Gain % = (130 / 1300) × 100 = 10% (B).

Q.38. Correct Option: (B)
Explanation: CP = 1350 / 0.90 = ■ 1,500. Required SP for 10% gain = 1500 × 1.10 = ■ 1,650 (B).

Q.39. Correct Option: (B)
Explanation: Total marks = 5 × 78 = 390. Sum of 4 students = 72 + 80 + 85 + 68 = 305. 5th student = 390 – 305 = 85 (B).

Q.40. Correct Option: (B)
Explanation: Failure percentage = 100% – 90% = 10%. Number of failed students = 10% of 450 = 45 (B).

Q.41. Correct Option: (B)
Explanation: SI = (12000 × 8 × 2.5) / 100 = ■ 2,400. Total Amount = 12,000 + 2,400 = ■ 14,400 (B).

Q.42. Correct Option: (B)
Explanation: Rate = (SI × 100) / (P × T) = (1440 × 100) / (6000 × 3) = 144 / 18 = 8% (B).

Q.43. Correct Option: (B)
Explanation: Let angle be x. x = 3(180° – x) => 4x = 540° => x = 135° (B).

Q.44. Correct Option: (B)
Explanation: Consecutive interior angles are supplementary. Other angle = 180° – 65° = 115° (B).

Q.45. Correct Option: (B)
Explanation: Sum of acute angles in right triangle = 90°. Other angle = 90° – 38° = 52° (B).

Q.46. Correct Option: (B)
Explanation: Side = 64 / 4 = 16 m. Area = 16^2 = 256 m^2 (B).

Q.47. Correct Option: (B)
Explanation: Area = 35 × 20 = 700 m^2. Cost = 700 × ■ 8 = ■ 5,600 (B).

Q.48. Correct Option: (B)
Explanation: Radius = 35 cm. Circumference = 2 × (22/7) × 35 = 220 cm = 2.2 m. Distance in 100 rev = 2.2 × 100 = 220 m (B).

Q.49. Correct Option: (B)
Explanation: 6 a^2 = 294 => a^2 = 49 => a = 7 cm. Volume = 7^3 = 343 cm^3 (B).

Q.50. Correct Option: (B)
Explanation: Volume = 3 × 2 × 1.5 = 9 m^3. Capacity = 9 × 1,000 = 9,000 Litres (B).

Q.51. Correct Option: (B)
Explanation: Time = Distance / Speed = 180 / 45 = 4 hours (B).

Q.52. Correct Option: (B)
Explanation: Speed = 400 / 50 = 8 m/s. In km/hr = 8 × (18/5) = 144 / 5 = 28.8 km/hr (B).

Q.53. Correct Option: (B)
Explanation: 4y – 12 = 2y + 10 => 2y = 22 => y = 11 (B).

Q.54. Correct Option: (B)
Explanation: (4^2 + 2^2) ÷ (3 + 1) = (16 + 4) ÷ 4 = 20 ÷ 4 = 5 (B).

Q.55. Correct Option: (B)
Explanation: x – y = 18, x + y = 84. Subtracting equations gives 2y = 66 => y = 33 (B).

Q.56. Correct Option: (B)
Explanation: Formula: F = (9/5)C + 32. Setting F = C = x gives x = –40° (B).

Q.57. Correct Option: (A)
Explanation: 28 – 42 + 15 – 11 = 43 – 53 = –10 (A).

Q.58. Correct Option: (B)
Explanation: Primes < 20: 2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 = 77 (B).

Q.59. Correct Option: (B)
Explanation: Last 3 digits of 7,816 are 816, which is divisible by 8 (816 / 8 = 102). Option (B) is correct.

Q.60. Correct Option: (B)
Explanation: Product of extremes = Product of means. 3 × 27 = 9 × x => 81 = 9x => x = 9 (B).

Q.61. Correct Option: (B)
Explanation: Perimeter of equilateral triangle = 3 × side = 3 × 14.2 cm = 42.6 cm (B).

Q.62. Correct Option: (B)
Explanation: 45 L = 45,000 mL. Number of bottles = 45,000 / 500 = 90 (B).

Q.63. Correct Option: (B)
Explanation: 482 rounds to 500; 319 rounds to 300. Estimated product = 500 × 300 = 1,50,000 (B).

Q.64. Correct Option: (A)
Explanation: An angle less than 90° is an Acute Angle (A).

Q.65. Correct Option: (B)
Explanation: A rectangle has 2 lines of symmetry (connecting midpoints of opposite sides) (B).

Q.66. Correct Option: (B)
Explanation: Total height = 4 × 145 = 580 cm. Sum of 3 girls = 440 cm. Fourth girl = 580 – 440 = 140 cm (B).

Q.67. Correct Option: (B)
Explanation: ■ 3.00 = 300 paise. Percentage = (45 / 300) × 100 = 15% (B).

Q.68. Correct Option: (B)
Explanation: The perimeter of a circle is called its Circumference (B).

Q.69. Correct Option: (C)
Explanation: Statement I is true (odd + odd = even). Statement II is false (3 × 5 = 15, which is composite) (C).

Q.70. Correct Option: (B)
Explanation: B = (HCF × LCM) / A = (12 × 180) / 36 = 60 (B).

Q.71. Correct Option: (B)
Explanation: Cost Price = 12 × 250 = ■ 3,000. Selling Price = ■ 3,600. Gain = 600. Gain % = (600 / 3000) × 100 = 20% (B).

Q.72. Correct Option: (B)
Explanation: LCM of 10, 5, 20 = 20. (14 + 8 – 3) / 20 = 19 / 20 (B).

Q.73. Correct Option: (B)
Explanation: 7x = 42 => x = 6. Son's age = 2 × 6 = 12 years (B).

Q.74. Correct Option: (B)
Explanation: Speed = 45 / 3 = 15 km/hr. Distance in 5 hours = 15 × 5 = 75 km (B).

Q.75. Correct Option: (B)
Explanation: Number of students = 4.5 × 20 = 90 students (B).

SOLUTIONS : SECTION C : INTELLIGENCE

Q.76. Correct Option: (B)
Explanation: Differences between consecutive terms: +4, +6, +8, +10, +12. Next term = 30 + 12 = 42 (B).

Q.77. Correct Option: (B)
Explanation: Pattern: -2 position decrement in alphabet (26, 24, 22, 20, 18, 16). 16th letter is P (B).

Q.78. Correct Option: (B)
Explanation: Doctor works in a Hospital; similarly, Teacher works in a School (B).

Q.79. Correct Option: (B)
Explanation: Pattern is n : n^2. $8^2 = 64$, so $11^2 = 121$ (B).

Q.80. Correct Option: (C)
Explanation: Dog, Cat, and Cow are domestic animals; Lion is a wild carnivore (C).

Q.81. Correct Option: (D)
Explanation: 49 (7^2), 81 (9^2), and 100 (10^2) are perfect squares; 125 is a perfect cube (5^3) (D).

Q.82. Correct Option: (A)
Explanation: Each letter shifts +1 forward: T+1=U, I+1=J, G+1=H, E+1=F, R+1=S -> UJHFS (A).

Q.83. Correct Option: (B)
Explanation: 'Only son of my father' is the man himself. Therefore, the woman is his Daughter (B).

Q.84. Correct Option: (B)
Explanation: Direct distance = $\sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$ km (B).

Q.85. Correct Option: (B)
Explanation: Position from left = Total – Right position + 1 = 30 – 15 + 1 = 16th (B).

Q.86. Correct Option: (A)
Explanation: Dictionary order: Package (2), Page (4), Pain (5), Palace (3), Paper (1) -> 2, 4, 5, 3, 1 (A).

Q.87. Correct Option: (A)
Explanation: Ascending order: Second (2) < Minute (4) < Hour (1) < Day (3) < Week (5) -> 2, 4, 1, 3, 5 (A).

Q.88. Correct Option: (B)
Explanation: Substitute symbols: 20 × 4 ÷ 2 – 10 + 5 = 20 × 2 – 10 + 5 = 40 – 10 + 5 = 35 (B).

Q.89. Correct Option: (A)
Explanation: All letters of ACTION (A, C, T, I, O, N) are present in COMPUTATION. Option (A) is correct.

Q.90. Correct Option: (B)
Explanation: Pattern: First × Second. Row 1: 5×4=20. Row 2: 6×7=42. Row 3: 8×9 = 72 (B).

Q.91. Correct Option: (A)
Explanation: Mirror image reverses digit sequence right-to-left and flips individual numbers horizontally (A).

Q.92. Correct Option: (A)
Explanation: Engineers and Doctors are distinct professionals, both enclosed in Human Beings (A).

Q.93. Correct Option: (B)
Explanation: 3×3 grid has: $3^2 + 2^2 + 1^2 = 9 + 4 + 1 = 14$ total squares (B).

Q.94. Correct Option: (B)
Explanation: Difference = 11 days. 11 ÷ 7 = 1 week + 4 odd days. Friday + 4 days = Tuesday (B).

Q.95. Correct Option: (D)
Explanation: Sun gives Light, Fire gives Heat, Ice gives Cold; Moon is a natural satellite, not a star (D).

Q.96. Correct Option: (B)
Explanation: Only 'Some roses may fade quickly' is logically valid without over-generalizing (B).

Q.97. Correct Option: (B)
Explanation: Boundary distance of a circle is perimeter; boundary of a polygon is also perimeter (B).

Q.98. Correct Option: (B)
Explanation: B=2, L=12, U=21, E=5. Sum = 2 + 12 + 21 + 5 = 40 (B).

Q.99. Correct Option: (A)
Explanation: Since P is to the immediate left of Q, Q must be to the immediate right of P (A).

Q.100. Correct Option: (B)
Explanation: Sum of opposite faces = 7. Face opposite to 3 = 7 – 3 = 4 (B).

SOLUTIONS : SECTION D : GENERAL KNOWLEDGE

Q.101. Correct Option: (B)
Explanation: A Barometer measures atmospheric air pressure (B).

Q.102. Correct Option: (B)
Explanation: The Ganges River Dolphin was declared India's National Aquatic Animal in 2009 (B).

Q.103. Correct Option: (B)
Explanation: Bharatanatyam is an ancient classical dance form of Tamil Nadu (B).

Q.104. Correct Option: (B)
Explanation: Major General in Army = Air Vice Marshal in Air Force = Rear Admiral in Navy (B).

Q.105. Correct Option: (B)
Explanation: Ultraviolet rays in sunlight trigger Vitamin D synthesis in human skin (B).

Q.106. Correct Option: (B)
Explanation: Deficiency of iodine causes enlargement of the thyroid gland, termed Goitre (B).

Q.107. Correct Option: (B)
Explanation: Snakes lack external ears and detect ground vibrations via their lower jaw and skin (B).

Q.108. Correct Option: (B)
Explanation: Insectivorous plants trap and digest insects to fulfill nitrogen requirements (B).

Q.109. Correct Option: (B)
Explanation: A Johad is a community-managed earthen check dam used to store rainwater in Rajasthan (B).

Q.110. Correct Option: (A)
Explanation: Pure water freezes into ice at 0°C (32°F) (A).

Q.111. Correct Option: (B)
Explanation: Jupiter is a gas giant and the largest planet in the Solar System (B).

Q.112. Correct Option: (B)
Explanation: The 13th-century Sun Temple is located at Konark, Odisha (B).

Q.113. Correct Option: (A)
Explanation: Maharshi Valmiki is revered as the Adi Kavi who authored the Sanskrit Ramayana (A).

Q.114. Correct Option: (B)
Explanation: INS Vikramaditya is a flagship aircraft carrier of the Indian Navy (B).

Q.115. Correct Option: (B)
Explanation: Wind energy is continuously replenished naturally and is renewable (B).

Q.116. Correct Option: (B)
Explanation: UNESCO headquarters is situated in Paris, France (B).

Q.117. Correct Option: (B)
Explanation: A baby goose is called a Gosling (B).

Q.118. Correct Option: (B)
Explanation: Evaporation is the process of liquid water converting into gas/vapour (B).

Q.119. Correct Option: (B)
Explanation: The Santosh Trophy is a premier state-level football championship in India (B).

Q.120. Correct Option: (B)
Explanation: Hemoglobin in Red Blood Cells (RBCs) binds and transports oxygen (B).

Q.121. Correct Option: (B)
Explanation: Abhinav Bindra won Gold in 10m Air Rifle shooting at 2008 Beijing Olympics (B).

Q.122. Correct Option: (B)
Explanation: Earth's gravity attracts all mass toward its center (Gravitational force) (B).

Q.123. Correct Option: (B)
Explanation: Assam produces more than half of India's total tea output (B).

Q.124. Correct Option: (B)
Explanation: Jim Corbett National Park is located in Nainital district, Uttarakhand (B).

Q.125. Correct Option: (B)
Explanation: Dr. Vikram Sarabhai initiated space research and helped establish ISRO (B).